Experiment No. 6

Write the Python program to implement BFS (Breadth First Search)

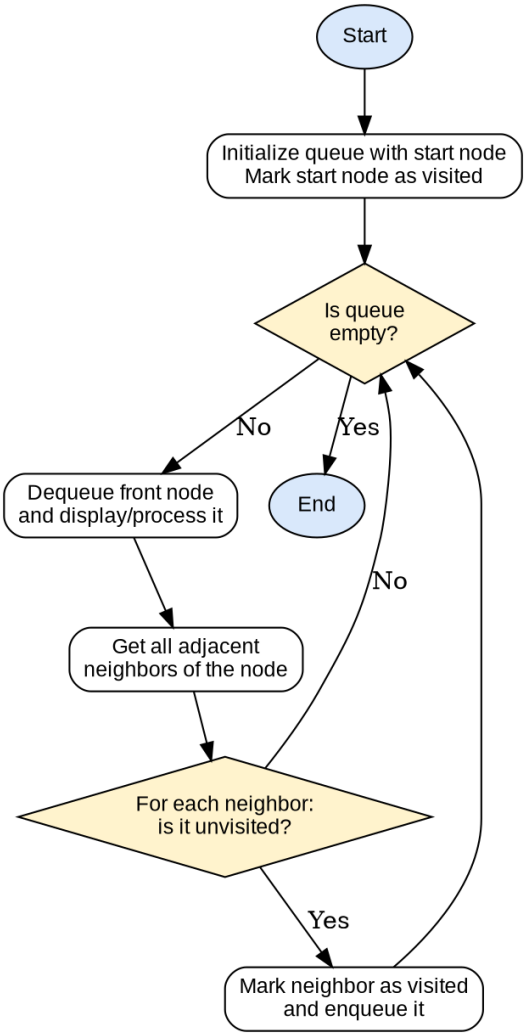
Aim

To write a Python program to implement the Breadth First Search (BFS) algorithm for traversing a graph.

Algorithm

1. Start.
2. Represent the graph using an adjacency list.
3. Initialize an empty queue and an empty visited set; add the start node to both.
4. Repeat while the queue is not empty:
5. (a) Dequeue the front node and add it to the traversal order.
6. (b) For every unvisited neighbor of this node, mark it visited and enqueue it.
7. When the queue becomes empty, the traversal order contains all nodes reachable from the start node.
8. Display the BFS traversal order.
9. Stop.

Flowchart



Program

```
from collections import deque

def bfs(graph, start):
    visited = set()
    queue = deque([start])
    visited.add(start)
    order = []

    while queue:
        node = queue.popleft()
        order.append(node)
        for neighbor in graph[node]:
            if neighbor not in visited:
```

```
                visited.add(neighbor)
                queue.append(neighbor)
    return order


graph = {
    'A': ['B', 'C'],
    'B': ['A', 'D', 'E'],
    'C': ['A', 'F'],
    'D': ['B'],
    'E': ['B', 'F'],
    'F': ['C', 'E']
}

start_node = 'A'
result = bfs(graph, start_node)
print("BFS Traversal starting from node", start_node, ":")
print(" -> ".join(result))
```

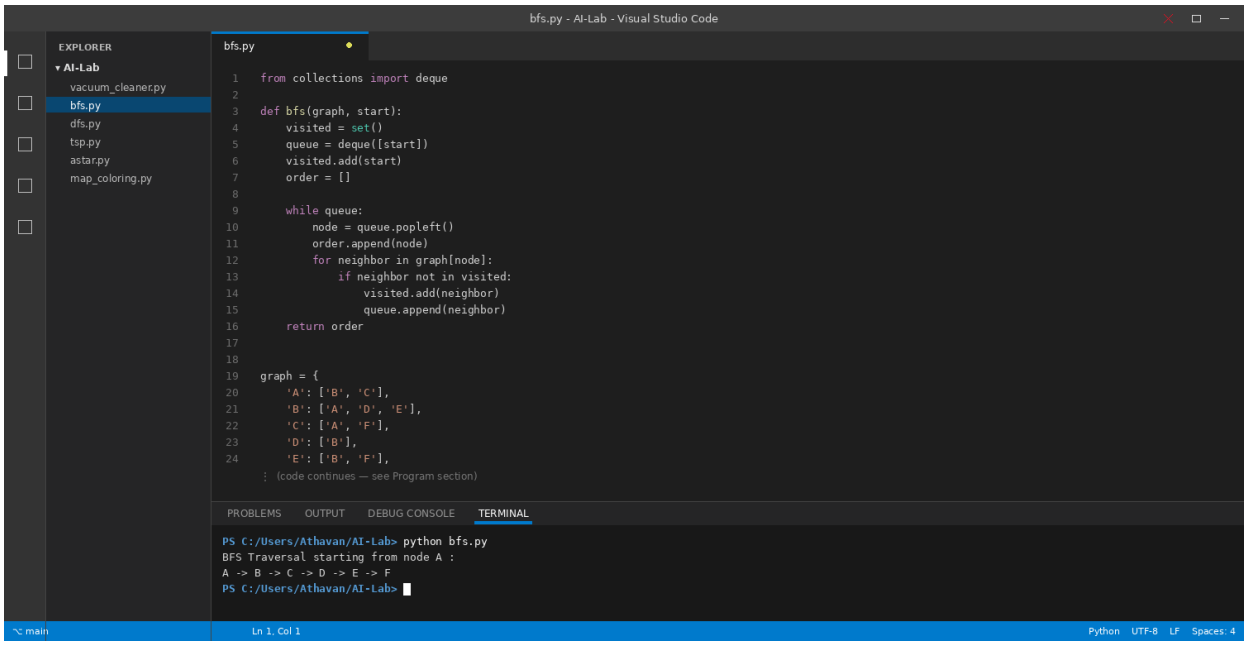
Result

The program was executed successfully and the output was verified as correct.

```
BFS Traversal starting from node A :
A -> B -> C -> D -> E -> F
```

Sample Output — Running in VS Code

Screenshot of the program being run in the Visual Studio Code integrated terminal:


bfs.py - AI-Lab - Visual Studio Code
EXPLORER
AI-Lab
vacuum_cleaner.py
bfs.py
dfs.py
tsp.py
astar.py
map_coloring.py
bfs.py
from collections import deque
def bfs(graph, start):
visited = set()
queue = deque([start])
visited.add(start)
order = []
while queue:
node = queue.popleft()
order.append(node)
for neighbor in graph[node]:
if neighbor not in visited:
visited.add(neighbor)
queue.append(neighbor)
return order
graph = {
'A': ['B', 'C'],
'B': ['A', 'D', 'E'],
'C': ['A', 'F'],
'D': ['B'],
'E': ['B', 'F'],
(code continues — see Program section)
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PS C:/Users/Athavan/AI-Lab> python bfs.py
BFS Traversal starting from node A :
A -> B -> C -> D -> E -> F
PS C:/Users/Athavan/AI-Lab>
main
Ln 1, Col 1
Python
UTF-8
LF
Spaces: 4